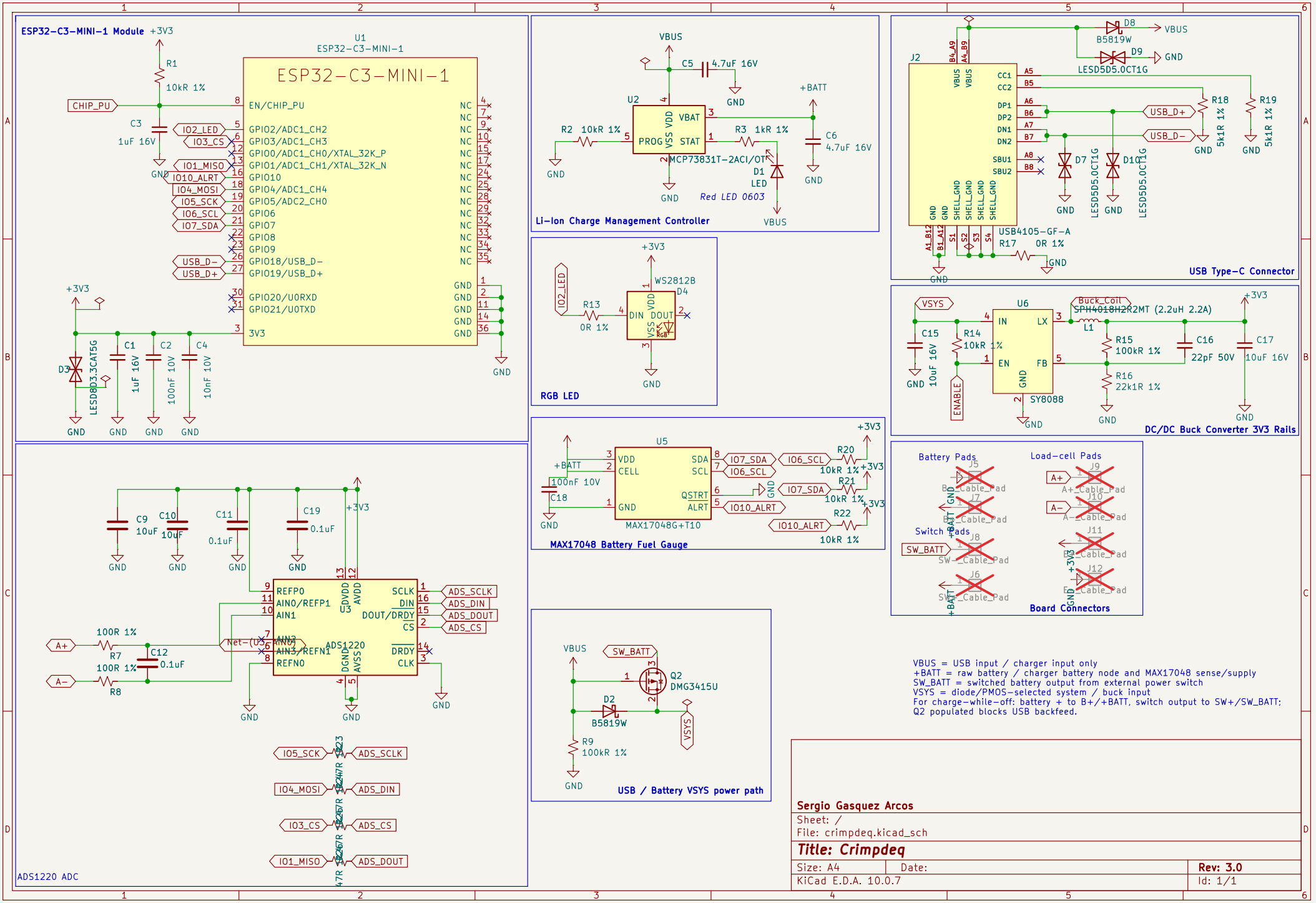

ESP32-C3-MINI-1 Module
U1
ESP32-C3-MINI-1
ESP32-C3-MINI-1
EN/CHIP_PU
GPIO2/ADC1_CH2
GPIO3/ADC1_CH3
GPIO0/ADC1_CH0/XTAL_32K_P
GPIO1/ADC1_CH1/XTAL_32K_N
GPIO10
GPIO4/ADC1_CH4
GPIO5/ADC2_CH0
GPIO6
GPIO7
GPIO8
GPIO9
GPIO18/USB_D-
GPIO19/USB_D+
GPIO20/U0RXD
GPIO21/U0TXD
3V3
NC
GND
CHIP_PU
IO2_LED
IO3_CS
IO1_MISO
IO10_ALRT
IO4_MOSI
IO5_SCK
IO6_SCL
IO7_SDA
USB_D-
USB_D+
R1 10kR 1%
C3 1uF 16V
D3 LESD8D3.3CAT5G
C1 1uF 16V
C2 100nF 10V
C4 10nF 10V
Li-Ion Charge Management Controller
U2 MCP73831T-2ACI/OT
C5 4.7uF 16V
C6 4.7uF 16V
R2 10kR 1%
R3 1kR 1%
D1 LED Red LED 0603
VBUS
+BATT
RGB LED
WS2812B
R13 0R 1%
USB Type-C Connector
J2
USB4105-GF-A
D8 B5819W
D9 LESD5D5.0CT1G
D7
D10
R17 0R 1%
R18 5k1R 1%
R19 5k1R 1%
DC/DC Buck Converter 3V3 Rails
U6 SY8088
Buck_Coil SPH4018H2R2MT (2.2uH 2.2A)
L1
VSYS
ENABLE
C15 10uF 16V
R14 10kR 1%
R15 100kR 1%
R16 22k1R 1%
C16 22pF 50V
C17 10uF 16V
MAX17048 Battery Fuel Gauge
U5 MAX17048G+T10
C18 100nF 10V
R20 10kR 1%
R21 10kR 1%
R22 10kR 1%
Battery Pads
Load-cell Pads
Switch Pads
Board Connectors
SW_BATT
ADS1220 ADC
U3 ADS1220
C9 10uF
C10 10uF
C11 0.1uF
C19 0.1uF
C12 0.1uF
R7 100R 1%
R8 100R 1%
ADS_SCLK
ADS_DIN
ADS_DOUT
ADS_CS
USB / Battery VSYS power path
Q2 DMG3415U
D2 B5819W
R9 100kR 1%
VBUS = USB input / charger input only
+BATT = raw battery / charger battery node and MAX17048 sense/supply
SW_BATT = switched battery output from external power switch
VSYS = diode/PMOS-selected system / buck input
For charge-while-off: battery + to B+/+BATT, switch output to SW+/SW_BATT;
Q2 populated blocks USB backfeed.
Sergio Gasquez Arcos
Sheet: /
File: crimpdeq.kicad_sch
Title: Crimpdeq
Size: A4
Date:
Rev: 3.0
KiCad E.D.A. 10.0.7
Id: 1/1